Preserving lesion quantification in longitudinal whole-body PSMA PET/CT registration

Fryderyk Kögl[a,b,c,*], **Anna Reithmeir**[a,b,c], **Daniel M. Lang**[a], **Emily Chan**[a,b], **Lukas Förner**[d], **Rickmer Braren**[e], **Julia A. Schnabel**[a,b,c,f]

[a]Helmholtz Munich, Neuherberg, Germany
[b]Technical University of Munich, Munich, Germany
[c]Munich Center for Machine Learning, Munich, Germany
[d]University Hospital Augsburg, Department of Diagnostic and Interventional Radiology and Neuroradiology, Augsburg, Germany
[e]University Medical Center Hamburg-Eppendorf, Hamburg, Germany
[f]King's College London, London, United Kingdom

Abstract. Purpose: [TODO: One or two sentences: longitudinal PSMA PET/CT registration should align anatomy without altering lesion volume and uptake.]
Approach: [TODO: LapIRN + instance optimisation with PET-preservation and bone-rigidity losses; patient-level split; comparison with NiftyReg and ConvexAdam; ablations of where the losses act.]
Results: [TODO: Main numbers after all experiments are in: accuracy vs. baselines, MTV/TLG change with and without the losses, IO-only placement, per-lesion effect for small bone lesions.]
Conclusions: [TODO: One or two sentences.]

Keywords: deformable image registration, PSMA PET/CT, longitudinal imaging, biomarker preservation, instance optimization.

*Fryderyk Kögl, fryderyk.koegl@tum.de

1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT is now routinely used for staging prostate cancer, for planning and following radioligand therapy, and for assessing response.[1] Patients therefore accumulate a baseline examination and one or more follow-up examinations, and registering each follow-up to its baseline makes it possible to compare tumour burden voxel by voxel, to track individual lesions, and to accumulate dose.[2] For these uses, a registration must do two things at once: it has to align the anatomy, and it must not itself change the PET-derived quantities that are compared across time points. Two such quantities are the metabolic tumour volume (MTV) and the total lesion glycolysis (TLG). A deformation that compresses or stretches a lesion changes both, so a change measured after registration can be an artefact of the registration rather than of the

disease. The Learn2Reg 2026 PSMAReg challenge[3–5] formalises this setting: each follow-up scan is warped to its baseline by a dense displacement field, which is scored on organ overlap, boundary distance, folding, and the change of MTV and TLG caused by the warp.

Classical registration solves an optimisation problem per image pair;[6–8] this is accurate but slow for whole-body scans. Learning-based registration replaces the per-pair optimisation with one forward pass of a trained network,[9–11] and instance optimisation (IO) can refine the network output per pair at test time. The coarse-to-fine Laplacian pyramid network LapIRN[12] recovers the large deformations between whole-body scans while keeping the transformation smooth and invertible.

In a challenge submission,[13] we adapted LapIRN to PSMA PET/CT and added losses that preserve the PET biomarkers of lesions and keep bones rigid. That submission was developed and evaluated within the challenge protocol. Here, we study these losses in a controlled setting: a new patient-level split with a held-out test set, retrained networks, tuned baselines, and statistics at the patient level. We ask two questions:

1. Do the PET-preservation and bone-rigidity losses improve the complete registration pipeline, and what do they cost in alignment accuracy?

2. Do the losses have to be part of network training, or is it sufficient to use them in instance optimisation?

[TODO: Contribution list once all results are in, e.g.: (i) a held-out evaluation of PET-preserving and rigidity losses for whole-body PSMA PET/CT registration, against tuned NiftyReg and ConvexAdam; (ii) the finding that applying the losses only in IO is as preserving and more accurate than training with them, while training with them is the only option without IO; (iii) a per-lesion

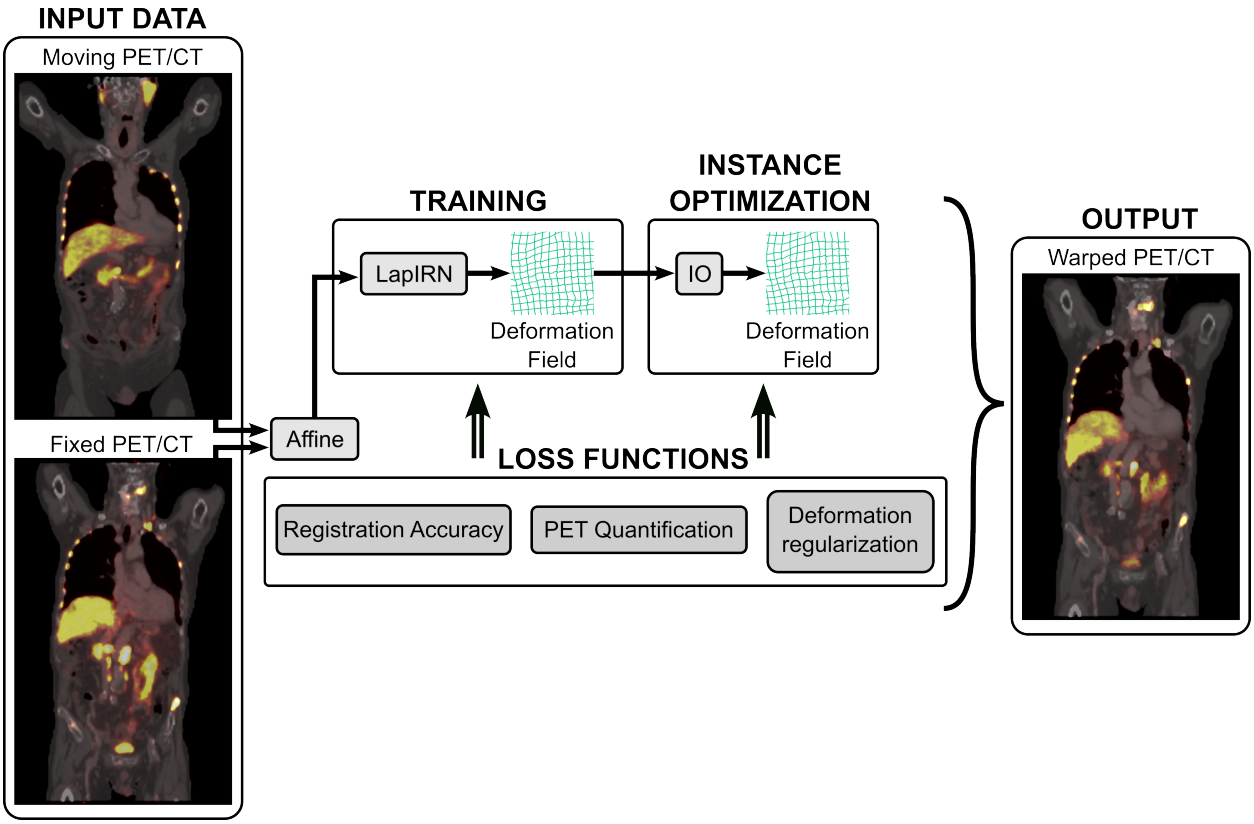


Fig 1 Registration pipeline. The follow-up (moving) and baseline (fixed) PET/CT scans are aligned by an affine pre-registration, then by the LapIRN network, which predicts a dense deformation. Instance optimisation (IO) refines this deformation per pair before the moving scan is resampled into the fixed frame. Network training and IO use the same groups of loss terms: alignment, PET preservation, and deformation regularisation (Sec. 2.3). [TODO: Update the figure if the rigidity term or the IO-only variant should appear.]

analysis showing where the losses act (small, bone-associated lesions); (iv) a comparison of existing rigidity penalties in low-resolution whole-body CT, including stencil leakage at bone boundaries.]

2 Methods

2.1 Data and Split

We used the training set of the PSMAReg challenge,[2,5] a longitudinal whole-body PSMA PET/CT cohort. Each scan consists of a CT and a PET image (in SUV), resampled to $2.7344 \times 2.7344 \times 3.27$ mm and cropped to $192 \times 192 \times 288$ voxels. For every scan, the challenge provides 117 CT

organ labels (the TotalSegmentator[14] classes) and a PSMA lesion mask. Following the challenge, a follow-up scan is always the moving image and the corresponding baseline scan the fixed image.

We split the 146 labelled patients with at least two scans at the patient level into training, validation, and test sets, aiming at 70/15/15 % of the registration pairs and balancing the sets for tumour burden and for bone- versus soft-tissue lesion location. Training uses all pairs of later onto earlier examinations of a patient; validation and testing use only follow-up-to-baseline pairs, as in the challenge. This gave 92/26/28 patients and 178/38/38 pairs. The split was built once with a fixed seed and frozen before any experiment of this study; all choices of settings, checkpoints, and methods were made on the validation set, and the test set was used only for the final evaluation. All patients were part of the challenge training data used to develop the earlier version of the method;[13] the test patients were not used for any decision in the present study.

Labels used by the method and for evaluation. Network training uses the provided organ and lesion labels of the training patients. Instance optimisation (Sec. 2.2) also needs organ and lesion labels, but of the scans being registered; to keep the evaluation independent of these inputs, IO uses only predicted labels, and the provided labels of validation and test scans are used only for evaluation (Sec. 3.1). Organ labels are predicted from the CT by TotalSegmentator.[14] Lesion masks are predicted from PET/CT by an nnU-Net[15] set up as the winning autoPET III solution:[16] a residual-encoder U-Net (ResEnc-L) with 192^3 patches, initialised from their pretrained weights, with an auxiliary organ-segmentation head and misalignment augmentation. It was trained with patient-grouped five-fold cross-validation on all 494 scans of the non-test patients, including patients with a single examination. Validation scans are segmented by the fold model that did not see the patient, and test scans by the ensemble of the five fold models. The predicted lesion masks

reach a mean Dice of 0.56 on the validation scans and 0.51 on the test scans (medians 0.69 and 0.67).

Preprocessing. For the network and IO, CT was clipped to $[-1000, 1500]$ HU and PET to $[0, 20]$ SUV, and both were rescaled to $[0, 1]$. As in the ConvexAdam baseline of the challenge,[17] the patient bed was removed from the CT. All PET quantities in the evaluation are computed on the original SUV images.

2.2 Registration Pipeline

The pipeline (Fig. 1) has three stages, all estimated from the moving scan (C_m, P_m) and the fixed scan (C_f, P_f), where C denotes CT and P PET. We denote the affine transform by φ_{aff}, the network deformation by φ_θ, and their composition, which maps the original moving scan to the fixed scan, by φ.

Affine pre-registration. ANTs[18] estimates an affine transform from the two CTs, using body masks, a $[-1000, 1000]$ HU window, half-resolution images, and Mattes mutual information. It is converted to a dense displacement field.

Deformable network. We use LapIRN[12] with three pyramid levels at quarter, half, and full resolution. Each level predicts a stationary velocity field that is added to the upsampled velocity of the coarser level and integrated by scaling and squaring[19] into φ_θ. Each level receives four channels: the fixed CT and PET and the affinely aligned moving CT and PET. The levels are trained one after the other; a finer level is initialised from the coarser one, which is first kept frozen and then

trained jointly. Each sub-network has 24 base channels and a trunk of ten residual blocks with an expansion factor of two.

Instance optimisation. At test time, the deformation of each pair is refined by optimising a residual stationary velocity field, initialised at zero and integrated by scaling and squaring (seven steps), on top of the total transformation φ. The residual is optimised at full resolution with Adam[20] (learning rate 0.0125, 40 iterations). The objective is the same as in training (Sec. 2.3), with the predicted labels in place of ground-truth labels. Of the 40 iterates, we return the one with the lowest objective, in which the interpolated (soft) MTV and TLG terms are replaced by their nearest-neighbour (hard) values, i.e. by how the evaluation counts them; this selection uses only predicted labels.

2.3 Registration Objective

Network training and IO minimise a weighted sum of three groups of terms:

$$\begin{aligned}\mathcal{L} = \underbrace{\lambda_{\mathrm{NCC}}\mathcal{L}_{\mathrm{NCC}} + \lambda_{\mathrm{DSC}}\mathcal{L}_{\mathrm{DSC}}}_{\text{alignment}} + \underbrace{\lambda_{\mathrm{MTV}}\mathcal{L}_{\mathrm{MTV}} + \lambda_{\overline{J}}\mathcal{L}_{\overline{J}} + \lambda_{J}\mathcal{L}_{J} + \lambda_{\mathrm{TLG}}\mathcal{L}_{\mathrm{TLG}}}_{\text{PET preservation}} \\ + \underbrace{\lambda_{\mathrm{rig}}\mathcal{L}_{\mathrm{rig}} + \lambda_{\mathrm{smooth}}\mathcal{L}_{\mathrm{smooth}} + \lambda_{\mathrm{NDV}}\mathcal{L}_{\mathrm{NDV}}}_{\text{regularisation}}\,.\end{aligned} \tag{1}$$

All terms are evaluated on the total transformation φ. The PET-preservation terms and the rigidity term are the subject of this study; they can be switched on in network training, in IO, or in both (Sec. 3). The two coarse network levels never use them, because lesions and thin bones span only a few voxels at these resolutions.

Alignment. $\mathcal{L}_{\mathrm{NCC}}$ is the negative local normalised cross-correlation of the CTs, summed over one to three scales depending on the pyramid level. PET is not used for similarity, because uptake changes between examinations with disease and therapy and a PET similarity would reward matching these changes. $\mathcal{L}_{\mathrm{DSC}}$ is a soft Dice loss on the CT organ labels.

PET preservation. Let M be the lesion mask of the moving scan and $\widetilde{M}$, $\widetilde{P}$ the linearly warped mask and PET. Nearest-neighbour warping, as used in the evaluation, is piecewise constant in the displacement and gives no useful gradient; linear warping gives a soft lesion boundary instead. The volume and uptake terms penalise the relative change caused by warping,

$$\mathcal{L}_{\mathrm{MTV}} = \left(\frac{\left| \sum_x \widetilde{M}(x) - \sum_x M(x) \right|}{\sum_x M(x) + \epsilon} \right)^2, \qquad \mathcal{L}_{\mathrm{TLG}} = \frac{\left| \sum_x \widetilde{P}(x)\widetilde{M}(x) - \sum_x P_m(x)M(x) \right|}{\sum_x P_m(x)M(x) + \epsilon}, \tag{2}$$

with $\epsilon = 10^{-5}$. Two further terms act on the Jacobian determinant $\det J_\varphi$ inside the warped lesion region W in the fixed frame:

$$\mathcal{L}_{\overline{J}} = \left(\frac{1}{|W|} \sum_{x \in W} \det J_\varphi(x) - 1 \right)^2, \qquad \mathcal{L}_J = \frac{1}{|W|} \sum_{x \in W} \big(\det J_\varphi(x) - 1 \big)^2. \tag{3}$$

$\mathcal{L}_{\overline{J}}$ constrains the net volume change of all lesions, and $\mathcal{L}_J$ additionally prevents compression in one part of a lesion from being compensated by expansion in another. For $\mathcal{L}_{\overline{J}}$, W is the linearly warped mask (without gradient through the mask), for $\mathcal{L}_J$ the nearest-neighbour warped mask.

Bone rigidity. Many PSMA lesions lie in bone, which moves rigidly between examinations. For each of the 61 skeletal labels of the fixed scan (in IO: the predicted labels), we fit a rigid transformation to the displacement of the structure in closed form[21,22] and penalise the mean squared

Table 1 Loss weights of Eq. (1) in network training (full-resolution level; the Dice weight is 3, 4, and 5 at the three levels) and in instance optimisation (IO).

	λ_{NCC}	λ_{DSC}	λ_{MTV}	$\lambda_{\overline{J}}$	λ_{J}	λ_{TLG}	λ_{rig}	$\lambda_{\mathrm{smooth}}$	λ_{NDV}
Training	5	5	5	0.125	1.25	1.25	0.05	10	10^4
IO	5	5	$4 \cdot 10^4$	10^3	4	16	0.25	0	10

residual of its voxels; $\mathcal{L}_{\mathrm{rig}}$ is the mean over structures with at least 50 voxels. Each bone is fitted independently, so neighbouring bones may move relative to each other, and the term involves no finite differences, so it does not constrain the tissue around a bone.

Regularisation. $\mathcal{L}_{\mathrm{smooth}}$ is the mean squared spatial gradient of the displacement. $\mathcal{L}_{\mathrm{NDV}}$ is a differentiable version of the challenge's non-diffeomorphic volume, which splits every voxel into tetrahedra and accumulates the negative part of their Jacobian determinants.[23]

Weights. Table 1 lists the weights. In training, the alignment weights are the same for all models, and the PET and rigidity weights are either used or set to zero. The IO weights were tuned on the validation set (Sec. 3.3).

2.4 Rigidity Penalties Compared

Besides the per-bone rigid fit, we evaluate the rigidity penalty of Staring et al.,[24] which requires the Jacobian J inside a bone to be a rotation: $\det J = 1$, $J^\top J = I$, and vanishing second derivatives of the deformation. It is computed with finite differences, whose stencil reads the 6 face and 12 edge neighbours of a voxel. At a bone boundary, part of the stencil lies outside the bone, so the penalty also acts on the surrounding soft tissue or on a neighbouring bone. A common remedy is to erode the bone mask so that the stencil stays inside; we erode each bone by exactly this 18-neighbour

footprint, which removes the leakage completely. The three variants (Staring, Staring with erosion, per-bone rigid fit) are compared in IO (Sec. 3).

2.5 Training

Each level was trained with Adam[20] for 10^5, 10^5, and $1.2 \cdot 10^5$ steps (quarter, half, full resolution) with learning rates of $1.5 \cdot 10^{-4}$, $1.2 \cdot 10^{-3}$, and 10^{-5}, a batch size of one with gradients accumulated over four pairs, a linear warm-up over five epochs, and a cosine decay to 10 % of the initial rate. A finer level is trained jointly with the coarser ones after ten epochs. Convolutions run in bfloat16, all losses in single precision. Training took 18 h, 22 h, and 3.7 days for the three levels on one NVIDIA A100 (80 GB). The augmentations mimic longitudinal variability: left–right flipping ($p = 0.5$), CT intensity shifts of ± 0.02 in the normalised window (about ± 50 HU), CT scaling between 0.9 and 1.1, PET scaling between 0.85 and 1.15, and field-of-view mismatch by cropping up to 40 axial slices from both scans and up to 10 further slices from each scan separately. The two coarse levels are shared by all models; the variants of this study differ only in the losses of the full-resolution level. Checkpoints were selected on the validation set: the coarse levels by CT Dice, the full-resolution level by the best accuracy (CT Dice and HD95) among checkpoints whose validation folding stays below $1.4 \cdot 10^{-4}$ % non-diffeomorphic volume. We did not select on MTV or TLG, so that the PET losses are not credited with effects of the checkpoint choice.

Use of AI tools. Parts of the code for training, evaluation, statistical analysis, and figure generation were written with the help of an AI coding assistant. All generated code was reviewed and tested by the authors, who are responsible for the analyses and results.

3 Experiments

3.1 Evaluation Metrics

All metrics are computed with the provided labels of the test scans, after warping the moving scan with the scorer of the challenge.

Alignment. CT Dice and the 95th percentile of the Hausdorff distance (HD95), averaged over the 117 organ labels.

PET preservation. As in the challenge, PET (SUV) is warped linearly and the lesion mask by nearest neighbour. MTV is the volume of the lesion mask and TLG the sum of SUV over it times the voxel volume; both are measured on the moving scan before and after warping. The error of $Q \in \{\mathrm{MTV}, \mathrm{TLG}\}$ is the absolute relative change,

$$|\Delta Q| = \frac{|Q_{\mathrm{after}} - Q_{\mathrm{before}}|}{\max(Q_{\mathrm{before}}, 10^{-6})}. \tag{4}$$

This measures what the registration does to the same scan; it does not assume that disease is unchanged between examinations. Because PET and mask are warped separately, the scorer changes TLG even for the identity transformation: on the test pairs with lesions, an all-zero displacement changes TLG by $2.9 \pm 1.2\,\%$ (MTV: $0\,\%$). This resampling floor applies to every warped method; we therefore report the unregistered scans ("Initial") without resampling.

Per-lesion change. The aggregate errors of Eq. (4) can hide large changes of individual lesions that cancel. We therefore also split the moving lesion mask into 26-connected components and compute $|\Delta\mathrm{MTV}|$ for each. Lesions are grouped by size (small < 0.5 ml, medium < 5 ml, large)

and location (bone-associated if at least half of the lesion lies on a skeletal label); lesions with fewer than five voxels are excluded.

Deformation quality. Percentage of non-diffeomorphic volume (NDV) inside the body, as defined by the challenge.[23]

Bone metrics. For the rigidity comparison, we additionally compute Dice and HD95 separately over bones and soft-tissue organs; the distortion of within-bone distances, i.e. the mean relative change of the physical distance between randomly sampled pairs of interior points (eroded mask) of the same bone, separately for short (5–15 mm) and long ($\geq$ 15 mm) distances; NDV inside bones and in a three-voxel band around them; and, as a secondary measure, the residual of a rigid fit to each bone's displacement (bone rigidity error). We also measure the fraction of stencil reads that leave the bone of the centre voxel (stencil leakage) and the fraction of bone voxels and of bones that remain after erosion.

3.2 Statistical Analysis

A patient may contribute several pairs and many lesions. We therefore average every metric over the pairs of a patient first and use patients as the unit of all tests, rankings, and confidence intervals (28 test patients). Paired differences between two methods are tested with the two-sided Wilcoxon signed-rank test[25] on patient values and corrected for multiple comparisons with Holm's method[26] within each table and metric; we call a difference significant if the corrected $p < 0.05$. Confidence intervals are 95 % percentile intervals over patient bootstrap samples;[27] for per-lesion statistics, patients are resampled together with all their lesions (cluster bootstrap).

For the comparison with baselines, we additionally rank all methods with the significance-based ranking of the challenge: per metric, every pair of methods is compared with a one-sided Wilcoxon test (Holm-corrected), and a method scores (number of methods it significantly outperforms $+1$)/K for K methods. Scores are averaged into accuracy (CT Dice, HD95), PET preservation ($|\Delta\mathrm{MTV}|$, $|\Delta\mathrm{TLG}|$), and regularity (NDV, after subtracting a practical-equivalence threshold of 0.005 % per pair), and combined into $100 \cdot \mathrm{acc}^{0.4}\,\mathrm{PET}^{0.4}\,\mathrm{reg}^{0.2}$. The challenge additionally scores Dice and HD95 of organs with physiological PET uptake, for which no labels are available to us; we omit them. Dice or HD95 of lesions would not be meaningful either, because lesions appear, disappear, and change between examinations. The stability of the ranks is shown by 95 % rank intervals over 1000 patient bootstrap samples.

3.3 Tuning and Model Selection

All tuning used the validation set and the challenge ranking above as the selection rule: the candidate settings of one grid are ranked against each other, and the top-ranked one is kept.

- **IO:** on the network trained with both losses, step-wise: first learning rate $\times$ iterations ($\{0.0125, 0.025, 0.05, 0.1\} \times \{10, 20, 40\}$), then a common scale of the four PET weights ($\times\{0, 0.1, 0.25, 0.5, 1, 2, 4\}$ of the training-derived values). The ranking does not reward rigidity (its cost in Dice is significant, while folding stays below the threshold for every weight), so it would set λ_{rig} to zero; we fixed $\lambda_{\mathrm{rig}} = 0.25$ instead, which costs 0.3 Dice points on validation.

- **NiftyReg**:[28,29] `reg_aladin` followed by `reg_f3d` on the GPU with normalised mutual information (local NCC is not available on the GPU), over the bending-energy weights

$\{0.0002, 0.0005, 0.001, 0.005, 0.02\}$ and final control-point spacings of $\{3, 5, 7, 9\}$ voxels; chosen: 0.0002 and 9 voxels.

- **ConvexAdam**:[17] after the same ANTs affine as our method, over the regularisation weight $\lambda \in \{0.1, 0.25, 0.5, 1.25, 2, 4\}$; chosen: 0.1.

Several choices ended at the edge of their grid; extending the grids changed the validation scores only marginally.

Choice of the full method. Before the experiments, we defined our full method as the network trained with both losses followed by IO with both losses. To check this choice, we also ranked all variants of Sec. 3.4 on the validation set. This ranking favoured the variant without any of the losses, because accuracy differences between variants are significant while differences in MTV and TLG rarely are. We therefore report both in the comparison with baselines, together with the variant that uses the losses only in IO.

3.4 Experiments

Effect of the losses. Four networks were trained with the same alignment and regularisation terms and with neither, the PET-preservation, the rigidity, or both groups of losses. Each is evaluated as network output and with IO using the same losses as in its training. As a reference for early stopping, the network without the losses is also evaluated at the checkpoint with the best validation tumour preservation.

Losses in training or only in IO. The network trained without the losses is combined with IO using PET preservation, rigidity, or both, and compared with the networks trained with these losses

(and IO with the same losses). To measure what the network contributes, IO with both losses is also run directly after the affine pre-registration.

Contribution of each PET term. On the network trained without the losses, IO with all four PET terms is compared with IO in which one term at a time is switched off.

Comparison with baselines. Initial alignment, the affine pre-registration, NiftyReg, Convex-Adam, IO after the affine, and our variants are compared on the test set. Runtimes are per pair on one NVIDIA RTX A6000 GPU and include the affine step.

Rigidity penalties. On the network trained with the PET losses, IO with the PET terms is combined with no rigidity, Staring, Staring with erosion, or the per-bone rigid fit. Because the penalties have different units, a reference weight is first calibrated for each so that its initial value on five validation pairs equals that of the rigid fit at $\lambda_{\text{rig}} = 0.25$ (Staring 0.66, Staring with erosion 1.03). Each method is then evaluated on the validation set at $\{0.1, 0.3, 1, 3, 10\}$ times its reference weight, giving a curve of CT Dice against within-bone distance distortion. For the test set, we choose for each method the largest weight whose validation Dice is within one Dice point of the variant without rigidity.

Accuracy–preservation trade-off. On the network trained without the losses, IO with the PET terms (without rigidity) is run with all four PET weights scaled by $\{0, 0.01, 0.03, 0.1, 0.3, 1\}$ relative to the training-derived values; $\times 0.1$ is the setting chosen by the IO tuning.

4 Results

All results are on the test set (28 patients, 38 pairs); values are mean±SD over patients, and differences are given in percentage points.

4.1 Effect of the PET-Preservation and Rigidity Losses

Table 2 shows the four networks trained with neither, the PET-preservation, the rigidity, or both groups of losses, as network output and with IO using the same losses. Training with the PET losses lowered the volume change of the lesions from 10.6 % to 7.6 % and the uptake change from 10.8 % to 7.6 %, at the cost of 1.9 Dice points. IO raised the Dice of every network by 2.8 to 4.4 points and kept the effect of the losses: with the PET losses, $|\Delta\mathrm{MTV}|$ was 6.3 % instead of 10.2 % and $|\Delta\mathrm{TLG}|$ 7.2 % instead of 11.9 %, for 2.8 Dice points. The rigidity loss alone did not change the Dice of the network and cost 0.5 points with IO, and it lowered $|\Delta\mathrm{MTV}|$ in both cases (network 9.4 %, with IO 7.9 %). Combining both groups preserved about as well as the PET losses alone and was the least accurate variant (72.8 % and 75.6 % Dice). HD95 followed the Dice in every case (network 7.2 to 7.5 mm, with IO 6.8 to 7.2 mm). No variant folded: the non-diffeomorphic volume of every pair stayed below the 0.005 % threshold of the challenge (at most 0.6 ppm on average).

Selecting the network without the losses at its checkpoint with the best validation tumour preservation, instead of the best accuracy, is no substitute for the losses: it cost 3.0 Dice points while $|\Delta\mathrm{MTV}|$ and $|\Delta\mathrm{TLG}|$ fell by only 0.4 and 0.9 points (not significant).

4.2 Losses in Training or Only in IO

Figure 2 compares, for each group of losses, the network trained with the losses (followed by IO with them) with the network trained without them, followed by the same IO. Using the losses only

Table 2 Effect of the PET and rigidity losses on the test set (28 patients, 38 pairs). Each network is trained with the given losses and evaluated as is (left) and with instance optimisation (IO) using the same losses (right). Mean±SD over patients. †/‡: significantly better/worse than "Neither" in the same block (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$).

	Network			Network + IO		
Losses	Dice (%) ↑	$\lvert\Delta\text{MTV}\rvert$ (%) ↓	$\lvert\Delta\text{TLG}\rvert$ (%) ↓	Dice (%) ↑	$\lvert\Delta\text{MTV}\rvert$ (%) ↓	$\lvert\Delta\text{TLG}\rvert$ (%) ↓
Neither	74.8±4.5	10.6±10.2	10.8±11.7	79.2±4.3	10.2±10.1	11.9±12.5
PET	72.9±4.4‡	7.6±9.0†	7.6±9.3†	76.4±4.7‡	6.3±9.1†	7.2±9.4†
Rigidity	74.8±4.5	9.4±9.5†	9.5±10.9†	78.7±4.2‡	7.9±9.0†	10.1±11.0
PET + rigidity	72.8±4.3‡	7.4±9.0†	7.7±10.1†	75.6±4.8‡	6.2±8.7†	7.9±9.0

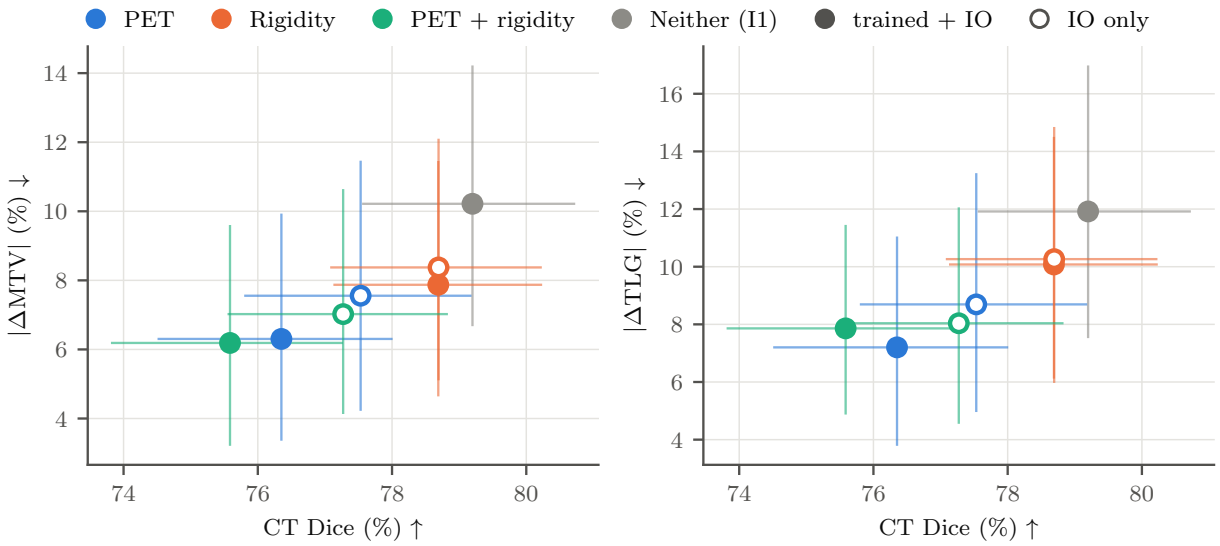


Fig 2 Losses in training and IO versus only in IO (test set, 28 patients). For each loss set, filled: network trained with the losses, then IO with them; hollow: network trained without them (T1), same IO. Grey: neither loss (I1). Points: patient means with 95 % patient-bootstrap confidence intervals.

in IO was more accurate: Dice increased by 1.2 points [95 % CI 0.4, 1.9] for the PET losses and by 1.7 points [1.1, 2.2] for both groups (both significant), and was unchanged for rigidity. Preservation was not significantly worse: $|\Delta\text{MTV}|$ increased by 1.3 [0.0, 2.6], 0.5 [0.0, 1.0], and 0.8 [-1.0, 2.7] points for PET, rigidity, and both, and $|\Delta\text{TLG}|$ by at most 1.5 points. Without IO, in contrast, the training losses are the only means of preservation (Table 2, left).

4.3 Per-Lesion Volume Change

Figure 3 shows the volume change of the 1327 individual lesions (26 patients with lesions). Small lesions changed most in every condition: for bone-associated lesions with IO and without the losses, the median change was 33 % (small), 17 % (medium), and 9 % (large). A small lesion has fewer than about 20 voxels, so gaining or losing one or two voxels is already a large relative change. Without the losses, IO increased the distortion of lesions, most of all of small bone lesions (median from 20 % to 33 %); bone lesions tended to grow (median signed change $+3$ to $+10$ %) and soft-tissue lesions to shrink (-9 to -13 %). The rigidity loss acted only on bone-associated lesions, as expected from its construction: with IO, it lowered the median change of small, medium, and large bone lesions from 33 % to 20 %, 17 % to 12 %, and 9 % to 5 %, and left soft-tissue lesions unchanged (17 to 22 %). The PET losses acted in both locations, for example from 18 % to 10 % for medium soft-tissue lesions and from 33 % to 13 % for small bone lesions; only for small soft-tissue lesions, the smallest group (41 lesions), no clear effect was visible. With both groups, bone lesions changed least at every size (10 %, 9 %, and 4 %), and their systematic growth disappeared (median signed change about 0). The medians of small lesions take only a few distinct values (e.g. 10.0 %, 12.5 %, 20.0 %) because they are ratios of small voxel counts.

4.4 Contribution of Each PET Term

[TODO: Leave-one-out table (tables/loo.tex) once L1–L4 have run.]

4.5 Comparison with Baselines

Table 3 and Fig. 4 compare our variants with the baselines. Every variant of ours was significantly more accurate than NiftyReg (69.4 % Dice, 8.6 mm HD95), ConvexAdam (61.7 %, 9.2 mm), and

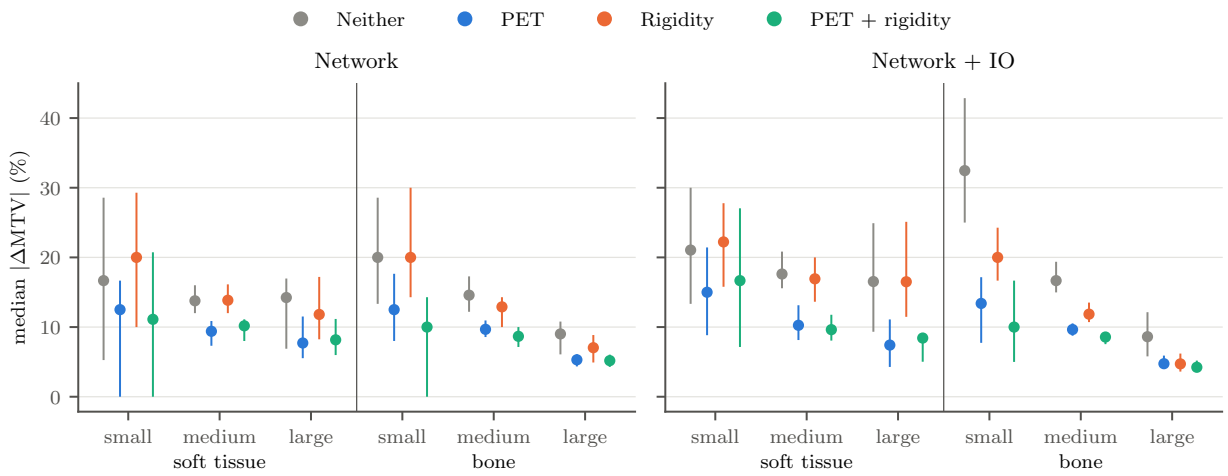


Fig 3 Per-lesion volume change on the test set (1327 lesions of 26 patients), by location (soft tissue / bone-associated, ≥50 % of the lesion on bone) and size (small <0.5 ml, medium <5 ml, large). Median $|\Delta\mathrm{MTV}|$ per lesion; left: network output, right: with instance optimisation (IO) using the same losses. Error bars: 95 % confidence intervals from a patient-cluster bootstrap. Lesions below 5 voxels are excluded.

the affine pre-registration (46.4 %, 13.9 mm), and none of them folded. NiftyReg, tuned on the validation set where it rarely folded, exceeded the folding threshold on 21 % of the test pairs (mean 1170 ppm, driven by three pairs with 0.8 to 1.3 %). IO directly after the affine pre-registration reached only 52.7 % Dice: within the same iteration budget, IO cannot replace the network. Its MTV and TLG change was the lowest of all methods (3.4 %), because the deformation it adds to the affine is small.

Among our variants, IO without the PET and rigidity losses was the most accurate (79.2 % Dice), and our full method, with the losses in training and IO, preserved best ($|\Delta\mathrm{MTV}|$ 6.2 %, significantly lower than without the losses). The variant with the losses only in IO ranked first (77.3 % Dice, 7.0 % $|\Delta\mathrm{MTV}|$), followed by the full method and the variant without the losses; their rank intervals overlap (1–5). The ranking was decided mainly by accuracy: after correction for multiple comparisons, few differences in preservation were significant (on $|\Delta\mathrm{MTV}|$, each variant with the losses outperformed one other method and IO after the affine two; on $|\Delta\mathrm{TLG}|$, only IO

Table 3 Comparison with baselines on the test set (28 patients, 38 follow-up→baseline pairs). Values are mean±SD over patients (a patient's pairs are averaged first). Rank: challenge significance ranking, with the 95 % interval of the rank over 1000 patient bootstrap samples in brackets, over all methods except Initial, which is a reference (its MTV/TLG change is 0 by construction because nothing is resampled). Training / IO: PET and rigidity losses (PET + rig.) used in training and in instance optimisation (IO); IO "–": no IO, "basic": IO with the similarity and regularisation terms only. †/‡: Ours + IO with PET + rig. in training and IO (our full method) is significantly better/worse than this method (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$; NDV tested after the 0.005 % equivalence threshold). $>$ thr.: pairs with %NDV above 0.005 %. Runtime per pair on one NVIDIA RTX A6000, including the affine step. Affine + IO: our instance optimisation directly after the affine pre-registration, without the network. Bold: best mean among ranked methods.

Method	Training	IO	Rank ↓	Dice (%) ↑	HD95 (mm) ↓	|ΔMTV| (%) ↓	|ΔTLG| (%) ↓	NDV (ppm) ↓	> thr. (%)	Runtime (s)
Initial	–	–	–	9.5±8.1†	44.37±21.32†	0.0±0.0	0.0±0.0	0±0	0	–
Affine	–	–	9 [8–9]	46.4±10.8†	13.92±4.74†	4.6±4.6	5.7±3.9	**0**±0	0	10.4±0.3
NiftyReg	–	–	7 [4–7]	69.4±4.7†	8.60±3.69†	7.6±12.5	11.3±13.7	1170±3120	21	107.7±24.9
ConvexAdam	–	–	8 [7–8]	61.7±4.5†	9.16±3.14†	9.3±12.0	9.6±15.8	**0**±0	0	11.3±0.6
Affine + IO	–	PET + rig.	5 [1–8]	52.7±12.1†	13.26±4.77†	**3.4**±3.2	**3.4**±3.6	0.1±0.099	0	155.5±2.6
Ours	–	–	6 [3–7]	74.8±4.5†	7.23±2.32	10.6±10.2†	10.8±11.7	0.0037±0.01	0	11.0±0.4
Ours	PET + rig.	–	4 [1–6]	72.8±4.3†	7.46±2.41†	7.4±9.0	7.7±10.1	0.0055±0.012	0	11.3±0.3
Ours + IO	–	basic	2 [1–5]	**79.2**±4.3‡	**6.78**±2.30‡	10.2±10.1†	11.9±12.5	0.58±0.26	0	153.9±2.6
Ours + IO	–	PET + rig.	1 [1–5]	77.3±4.5‡	7.01±2.33‡	7.0±8.8	8.0±10.2	0.25±0.25	0	157.2±2.6
Ours + IO	PET + rig.	PET + rig.	2 [1–5]	75.6±4.8	7.24±2.38	6.2±8.7	7.9±9.0	0.084±0.089	0	156.6±2.7

after the affine outperformed others), and none in folding. The network alone needs 11 s per pair, as fast as ConvexAdam and ten times faster than NiftyReg (108 s); IO adds about 2.5 minutes. Figure 5 shows a test pair. [TODO: Describe the qualitative example once the case is fixed (CASE in figure_qualitative.py).]

4.6 Rigidity Penalties

[TODO: After the validation sweeps and test runs: curve of Dice vs. distance distortion (validation), test table with bone/soft-tissue alignment, distance distortion, NDV in bone and boundary band, stencil leakage, erosion retention. Leakage on all 560 scans: 21.2 % of stencil reads leave the bone; erosion keeps 35.5 % of bone voxels and removes 23 % of bones (ribs: 2.4 % kept, 58 % removed).]

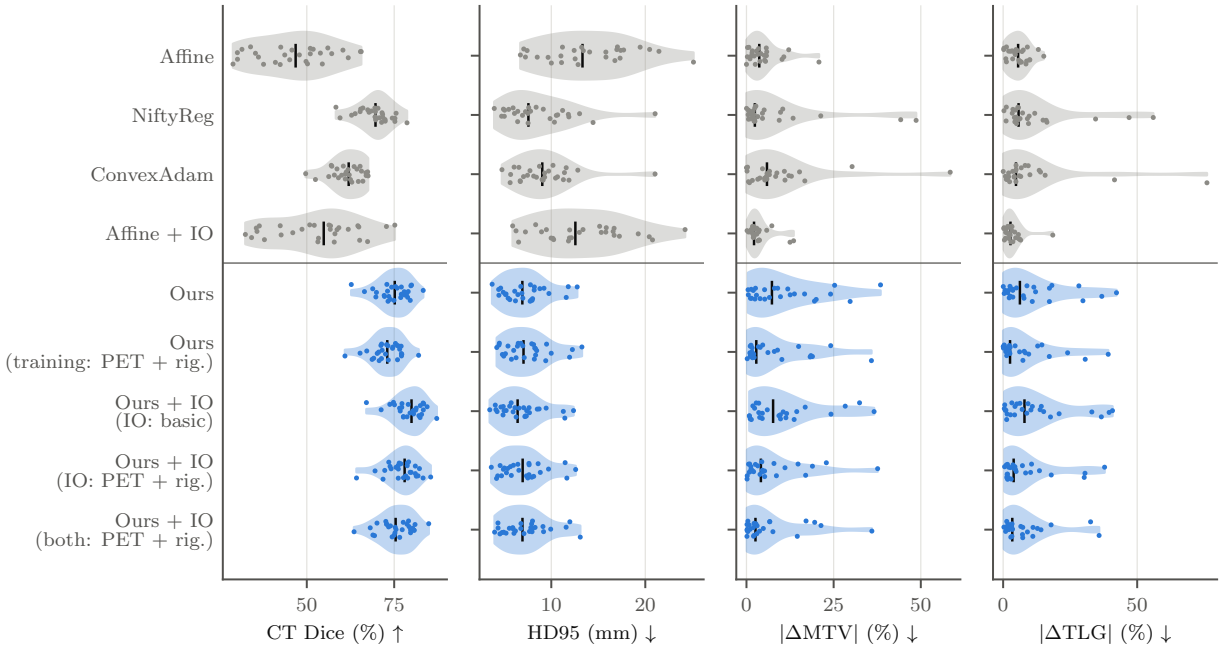


Fig 4 Patient-level results on the test set (28 patients; a patient's pairs are averaged). Violins: kernel density over patients, line: median; each dot is one patient. In brackets: where the PET and rigidity losses (PET + rig.) act; basic: IO with the similarity and regularisation terms only; both: training and IO. Initial alignment is omitted (CT Dice 9.5 %, HD95 44.4 mm; MTV/TLG change 0 by construction, Table 3).

4.7 Accuracy–Preservation Trade-off

[TODO: Trade-off curve (PET weight $\times\{0, 0.01, 0.03, 0.1, 0.3, 1\}$) once the runs are done; mark the tuned operating point.]

5 Discussion

[TODO: Write after all results are in. Points (findings.md):

- Main answer: the losses preserve lesion volume and uptake at a known accuracy cost; IO-only placement is as preserving and more accurate; without IO (about 14 times faster), training with the losses is the only way to preserve.
- Where the losses act: small and bone-associated lesions; rigidity only on bone lesions.

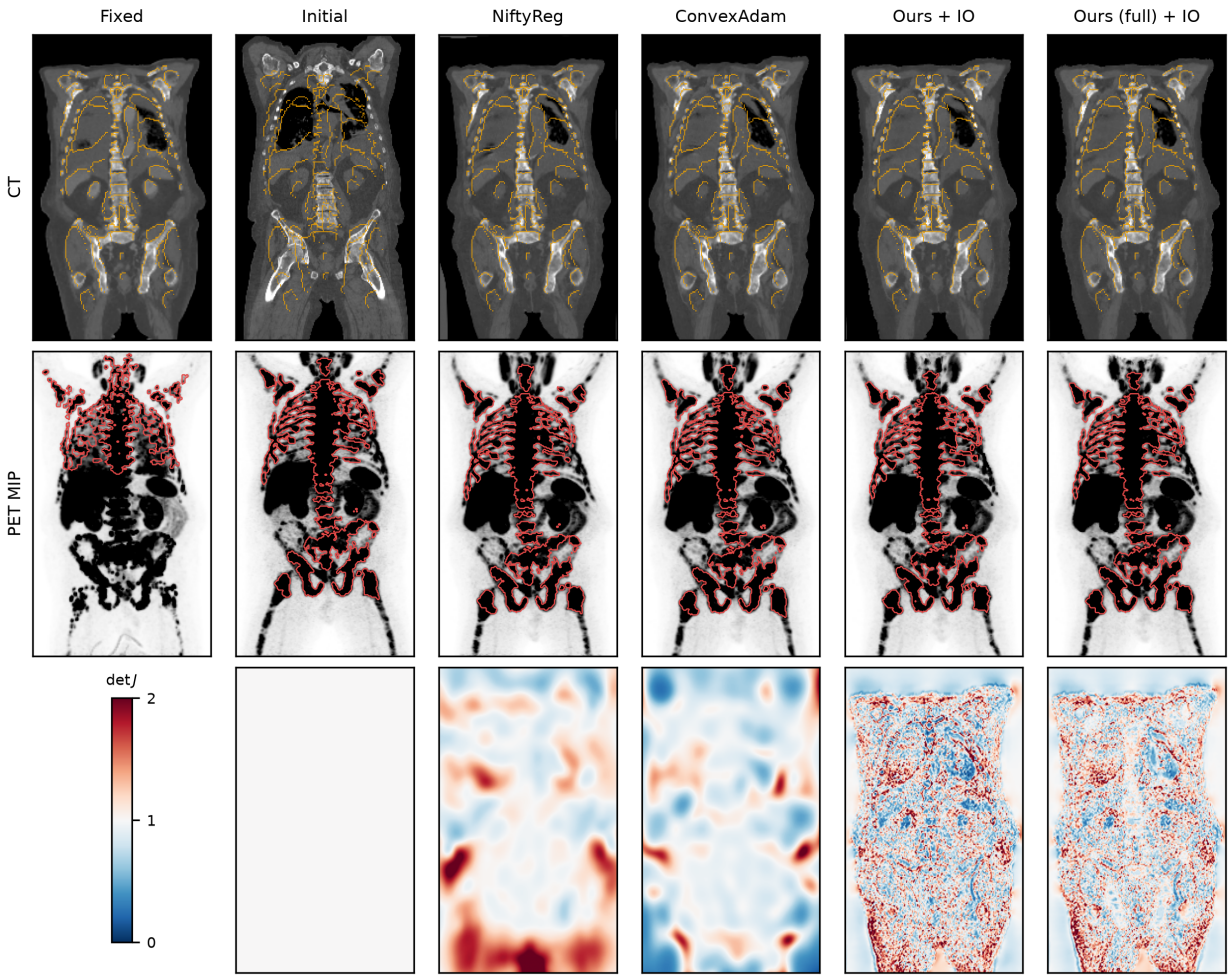


Fig 5 Test pair (patient 0054, follow-up 01 → baseline 00), chosen as the pair with at least 5 lesions whose CT Dice for Ours + IO is closest to the median. Ours (full): with the PET and rigidity losses in training and IO. Top: CT, coronal slice through the spine, with the fixed scan's organ-label boundaries (yellow). Middle: PET maximum-intensity projection (SUV 0–8) with the warped lesion mask (red); left: fixed scan and its lesions. Bottom: Jacobian determinant on the same slice (folds in black). Initial: the moving scan, not resampled.

- The challenge ranking is dominated by accuracy: biomarker differences rarely reach significance, and methods that barely deform score well on preservation (affine + IO). This also explains the validation choice of the variant without losses.

- TLG resampling floor of the scorer (2.9 % for the identity).

- PET losses in IO are limited by predicted lesion masks: with ground-truth masks the MTV change drops further (validation); a recall-biased mask made it worse.

- NiftyReg folds on a few test pairs although its validation folding was low.
- Rigidity: stencil leakage of finite-difference penalties at bone boundaries; erosion removes most of the skeleton at this resolution.
- Limitations: one cohort, low resolution, predicted labels, all patients seen during the development of the earlier version.

]

6 Conclusion

[TODO: Two or three sentences.]

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code, Data, and Materials Availability

All experiments use the publicly available training data of the Learn2Reg 2026 PSMAReg challenge,[3,4] which is adapted from the PSMA-PET-CT-Lesions dataset in The Cancer Imaging Archive[5] (DOI: 10.7937/r7ep-3x37). The private test data of the challenge were not used. The patient-level split into training, validation, and test sets is provided with the code. [TODO: Code repository link (training, IO, evaluation, split, and trained weights).]

Acknowledgments

F.K. and J.A.S. are partially supported by the German Federal Ministry of Research, Technology and Space (DECIPHER-M, 01KD2420G). A.R. is supported by funding from the Free State of Bavaria (StMGP) as part of the project "KI-gestützte multi-modale Diagnostik und stratifizierte Therapie für Endometriose (EndoKI)". D.M.L. and J.A.S. received funding from Helmholtz Imaging, a platform of the Helmholtz Information & Data Science Incubator. [TODO: Funding E.C.] L.F. was partially funded by the Intramural Research Funding Grant "AI-driven Longitudinal Lesion Tracking" of the Faculty of Medicine, University of Augsburg, by the Bavarian Center for Cancer Research as part of the Lighthouse "Local Therapies", and by the Bavarian Ministry of Economic Affairs, Regional Development and Energy (StMWi) under grant number DIK-2310-0004//DIK0556/02. An AI assistant was used for language editing of the manuscript; the authors reviewed and take responsibility for all content. The use of AI tools for code generation is described in Sec. 2.5.

References

1 Wolfgang P. Fendler, Matthias Eiber, Mohsen Beheshti ... Ken Herrmann, "68Ga-PSMA PET/CT: Joint EANM and SNMMI procedure guideline for prostate cancer imaging: version 1.0," *European Journal of Nuclear Medicine and Molecular Imaging* **44**, 1014–1024 (2017).

2 Learn2Reg Consortium, "Learn2Reg 2026 – PSMAReg: Registration of pre-therapy and follow-up whole-body PSMA PET/CT scans." Codabench challenge (2026). https://www.codabench.org/competitions/15724/.

3 Lasse Hansen, Wiebke Heyer, Christoph Großbröhmer ... Mattias P. Heinrich, "Learn2reg

2024: New benchmark datasets driving progress on new challenges," *Machine Learning for Biomedical Imaging* **3**, 775–791 (2025).

4 Alessa Hering, Lasse Hansen, Tony C. W. Mok ... Mattias P. Heinrich, "Learn2reg: Comprehensive multi-task medical image registration challenge, dataset and evaluation in the era of deep learning," *IEEE Transactions on Medical Imaging* **42**(3), 697–712 (2023).

5 Katharina Jeblick, Stefan O. Schönberg, and others, "A whole-body PSMA-PET/CT dataset with manually annotated tumor lesions (PSMA-PET-CT-Lesions)." The Cancer Imaging Archive (2024). Version 1.

6 Tom Vercauteren, Xavier Pennec, Aymeric Perchant ... Nicholas Ayache, "Diffeomorphic demons: Efficient non-parametric image registration," *NeuroImage* **45**(1, Supplement 1), S61–S72 (2009).

7 B.B. Avants, C.L. Epstein, M. Grossman ... J.C. Gee, "Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain," *Medical Image Analysis* **12**(1), 26–41 (2008). Special Issue on The Third International Workshop on Biomedical Image Registration - WBIR 2006.

8 D. Rueckert, L.I. Sonoda, C. Hayes ... D.J. Hawkes, "Nonrigid registration using free-form deformations: application to breast mr images," *IEEE Transactions on Medical Imaging* **18**(8), 712–721 (1999).

9 Guha Balakrishnan, Amy Zhao, Mert R. Sabuncu ... Adrian V. Dalca, "An unsupervised learning model for deformable medical image registration," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, (2018).

10 Adrian V. Dalca, Guha Balakrishnan, John Guttag ... Mert R. Sabuncu, "Unsupervised learn-

ing for fast probabilistic diffeomorphic registration," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*, Alejandro F. Frangi, Julia A. Schnabel, Christos Davatzikos ... Gabor Fichtinger, Eds., 729–738, Springer International Publishing, (Cham) (2018).

11 Junyu Chen, Eric C. Frey, Yufan He ... Yong Du, "Transmorph: Transformer for unsupervised medical image registration," *Medical Image Analysis* **82**, 102615 (2022).

12 Tony C. W. Mok and Albert C. S. Chung, "Large deformation diffeomorphic image registration with laplacian pyramid networks," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*, Anne L. Martel, Purang Abolmaesumi, Danail Stoyanov ... Leo Joskowicz, Eds., 211–221, Springer International Publishing, (Cham) (2020).

13 Fryderyk Kögl, Anna Reithmeir, Daniel M. Lang ... Julia A. Schnabel, "PET quantification-preserving registration for longitudinal whole-body PSMA PET/CT," (2026). arXiv:2610.21451.

14 Jakob Wasserthal, Hanns-Christian Breit, Manfred T. Meyer ... Martin Segeroth, "Totalsegmentator: Robust segmentation of 104 anatomic structures in CT images," *Radiology: Artificial Intelligence* **5**(5), e230024 (2023).

15 Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl ... Klaus H. Maier-Hein, "nnu-net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods* **18**, 203–211 (2021).

16 Maximilian Rokuss, Balint Kovacs, Yannick Kirchhoff ... Fabian Isensee, "From fdg to psma: A hitchhiker's guide to multitracer, multicenter lesion segmentation in pet/ct imaging," *ArXiv* (2024).

17 Hanna Siebert, Lasse Hansen, and Mattias P. Heinrich, "Fast 3d registration with accurate optimisation and little learning for learn2reg 2021," in *Biomedical Image Registration, Domain Generalisation and Out-of-Distribution Analysis*, Marc Aubreville, David Zimmerer, and Mattias Heinrich, Eds., 174–179, Springer International Publishing, (Cham) (2022).

18 Brian B. Avants, Nicholas J. Tustison, Gang Song ... James C. Gee, "A reproducible evaluation of ants similarity metric performance in brain image registration," *NeuroImage* **54**(3), 2033–2044 (2011).

19 Vincent Arsigny, Olivier Commowick, Xavier Pennec ... Nicholas Ayache, "A log-Euclidean framework for statistics on diffeomorphisms," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, *Lecture Notes in Computer Science* **4190**, 924–931, Springer (2006).

20 Diederik P. Kingma and Jimmy Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, (2015).

21 Wolfgang Kabsch, "A solution for the best rotation to relate two sets of vectors," *Foundations of Crystallography* **32**(5), 922–923 (1976).

22 S. Umeyama, "Least-squares estimation of transformation parameters between two point patterns," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **13**(4), 376–380 (1991).

23 Yihao Liu, Junyu Chen, Shuwen Wei ... Jerry Prince, "On finite difference jacobian computation in deformable image registration," *International Journal of Computer Vision* **132**, 3678–3688 (2024).

24 Marius Staring, Stefan Klein, and Josien P. W. Pluim, "A rigidity penalty term for nonrigid registration," *Medical Physics* **34**(11), 4098–4108 (2007).

25 Frank Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin* **1**(6), 80–83 (1945).

26 Sture Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics* **6**(2), 65–70 (1979).

27 Bradley Efron and Robert J. Tibshirani, *An Introduction to the Bootstrap*, Chapman & Hall, New York (1993).

28 Marc Modat, Gerard R. Ridgway, Zeike A. Taylor . . . Sébastien Ourselin, "Fast free-form deformation using graphics processing units," *Computer Methods and Programs in Biomedicine* **98**(3), 278–284 (2010). HP-MICCAI 2008.

29 Marc Modat, David M. Cash, Pankaj Daga . . . Sébastien Ourselin, "Global image registration using a symmetric block-matching approach," *Journal of Medical Imaging* **1**(2), 024003 (2014).

List of Figures

1 Registration pipeline. The follow-up (moving) and baseline (fixed) PET/CT scans are aligned by an affine pre-registration, then by the LapIRN network, which predicts a dense deformation. Instance optimisation (IO) refines this deformation per pair before the moving scan is resampled into the fixed frame. Network training and IO use the same groups of loss terms: alignment, PET preservation, and deformation regularisation (Sec. 2.3). [TODO: Update the figure if the rigidity term or the IO-only variant should appear.]

2 Losses in training and IO versus only in IO (test set, 28 patients). For each loss set, filled: network trained with the losses, then IO with them; hollow: network trained without them (T1), same IO. Grey: neither loss (I1). Points: patient means with 95 % patient-bootstrap confidence intervals.

3 Per-lesion volume change on the test set (1327 lesions of 26 patients), by location (soft tissue / bone-associated, $\geq$50 % of the lesion on bone) and size (small $<$0.5 ml, medium $<$5 ml, large). Median $|\Delta\mathrm{MTV}|$ per lesion; left: network output, right: with instance optimisation (IO) using the same losses. Error bars: 95 % confidence intervals from a patient-cluster bootstrap. Lesions below 5 voxels are excluded.

4 Patient-level results on the test set (28 patients; a patient's pairs are averaged). Violins: kernel density over patients, line: median; each dot is one patient. In brackets: where the PET and rigidity losses (PET + rig.) act; basic: IO with the similarity and regularisation terms only; both: training and IO. Initial alignment is omitted (CT Dice 9.5 %, HD95 44.4 mm; MTV/TLG change 0 by construction, Table 3).

5 Test pair (patient 0054, follow-up 01 $\rightarrow$ baseline 00), chosen as the pair with at least 5 lesions whose CT Dice for Ours + IO is closest to the median. Ours (full): with the PET and rigidity losses in training and IO. Top: CT, coronal slice through the spine, with the fixed scan's organ-label boundaries (yellow). Middle: PET maximum-intensity projection (SUV 0–8) with the warped lesion mask (red); left: fixed scan and its lesions. Bottom: Jacobian determinant on the same slice (folds in black). Initial: the moving scan, not resampled.

List of Tables

1 Loss weights of Eq. (1) in network training (full-resolution level; the Dice weight is 3, 4, and 5 at the three levels) and in instance optimisation (IO).

2 Effect of the PET and rigidity losses on the test set (28 patients, 38 pairs). Each network is trained with the given losses and evaluated as is (left) and with instance optimisation (IO) using the same losses (right). Mean$\pm$SD over patients. †/‡: significantly better/worse than "Neither" in the same block (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$).

3 Comparison with baselines on the test set (28 patients, 38 follow-up→baseline pairs). Values are mean±SD over patients (a patient's pairs are averaged first). Rank: challenge significance ranking, with the 95 % interval of the rank over 1000 patient bootstrap samples in brackets, over all methods except Initial, which is a reference (its MTV/TLG change is 0 by construction because nothing is resampled). Training / IO: PET and rigidity losses (PET + rig.) used in training and in instance optimisation (IO); IO "–": no IO, "basic": IO with the similarity and regularisation terms only. †/‡: Ours + IO with PET + rig. in training and IO (our full method) is significantly better/worse than this method (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$; NDV tested after the 0.005 % equivalence threshold). $>$ thr.: pairs with %NDV above 0.005 %. Runtime per pair on one NVIDIA RTX A6000, including the affine step. Affine + IO: our instance optimisation directly after the affine pre-registration, without the network. Bold: best mean among ranked methods.